Last class

1. bdd linear fn(als)
2. Homeomorphism
3. Closed operators $\Leftarrow$

X , Y

$T: X_0 \xrightarrow{\subseteq X} Y$ linear

& $x_n \to x$, $x_n \in X_0$ $x \in X$

$Tx_n \to y$ in Y.

Then $x \in X_0$ & $Tx = y$.

$\iff$ T is a closed operator.

Ex. bdd op $\not\Rightarrow$ closed.

$I: C^1[0,1] \subseteq C[0,1] \to C[0,1]$

closed op $\not\Rightarrow$ bdd.

$\subseteq C[0,1]$

$Tx = x'$ $\quad T : C^1[0,1] \to C[0,1]$

(seq of fns)

Qn Why we call these ops as closed op?

$$T : X_0 \subseteq X \longrightarrow Y$$

$$\text{Graph}(T) = Gr(T) \subseteq X \times Y$$

$$Gr(T) = \{ (x, Tx) \mid x \in X_0 \}$$

$$\|(x, Tx)\|_{Gr} = \|x\|_X + \|Tx\|_Y$$

- $Gr(T)$ is a nls.

Theo: A linear operator $T : X_0 \subseteq X \to Y$ is a closed operator $\Longleftrightarrow$ $Gr(T)$ is a closed subset of $X \times Y$.

Proof: Assume T is a closed operator.

Claim: $Gr(T)$ is closed.

Let (x_n, y_n) be a seq in $Gr(T)$.

s.t $(x_n, y_n) \to (x, y)$ in $Gr(T)$

Claim $(x, y) \in Gr(T)$.

$$\|(x_n, y_n) - (x, y)\|_{Gr(T)} = \underbrace{\|x_n - x\|_X}_{\to 0} + \underbrace{\|Tx_n - y\|_Y}_{\to 0}$$

$\to 0$ as $n \to \infty$

$\therefore$ $x_n \to x$ in X

& $Tx_n \to y$ in Y.

$\Rightarrow x \in X_0$ & $Tx = y$.

$(x, y) = (x, Tx) \in Gr(T)$.

$\therefore$ $Gr(T)$ is closed.

Let us assume $Gr(T)$ is closed.

Claim: T is closed.

Given $Gr(T)$ is closed

so a seq: $(x_n, Tx_n) \to (x, Tx) \in Gr(T)$

so $(x_n) \in X_0$

& $x \in X_0$ as $Gr(T)$ is closed.

T is closed means: $(x_n) \to x \in X$

$\in X_0$

st: $Tx_n \to y$

implies $x \in X_0$ & $Tx = y$

This can be concluded from $Gr(T)$ closed

Theo. Let $T : X \to Y$ be an injective bdd linear op.

Then $T^{-1} : \underset{\subseteq Y}{R(T)} \to X$

is a closed op.

Proof. Given T is an injective bdd linear op.

$\therefore$ T^{-1} exist from $R(T) \overset{\subseteq Y}{\to} X$.

Let $y_n \to y$ in Y.

& $T^{-1}(y_n) \to x$ in X. $\Big\}$ (1)

Then claim $T^{-1}(y) = x$.

or. $Tx = y$.

Let $x_n = T^{-1}(y_n)$, $x_n \in X$.

By ① $x_n \to x$ in X

& $Tx_n \to y$ in Y.

Now given T is bdd so

$Tx_n \to Tx$ in Y as $x_n \to x$ as $n \to \infty$.

$\therefore$ $Tx = y$. $\Rightarrow y \in R(T)$

& $T^{-1}y = x$.

$\therefore$ T^{-1} is a closed linear operator.

Qn. When a bdd op is closed?

bdd $\not\Rightarrow$ closed
closed $\not\Rightarrow$ bdd

Theo. Let $T: X_0 \to Y$ be a bdd operator. If X_0 is a closed subspace of X

then T is closed.

Theo.

Let $T : X_0 \subseteq X \to Y$ be a bdd op. If Y is a Banach space & T is closed then X_0 is closed.

Consequence.

Let X, Y be Banach space

& $X_0 \underset{\neq}{\subset} X$, dense.

Suppose $T : X_0 \subseteq X \to Y$ be closed.

Qn: Can T possesses an extension $\tilde{T} : X \underset{=\overline{X_0}}{} \to Y$ which is closed linear operator?

No.

Closed Graph Theorem

Later

If both the domain & codomain of a closed op are Banach then the op is bdd.

Suppos a closed linear ex. exists.

$$\tilde{T} : X \to Y.$$

By closed gr. Theo $\tilde{T}$ is bdd.

$\Rightarrow$ $T : X_0 \subseteq X \to Y$ is also bdd.

$\Rightarrow$ X_0 is closed ?

$\Downarrow$

since X_0 is dense.

Contrast to the bdd lin operator.

—

Theo. Suppose $T : \underline{X_0} \subseteq X \to Y$ is a closed op.

Then we have

i) $N(T)$ = null space of T

is closed.

2) If T is injective then
$$T^{-1} : R(T) \subseteq Y \to X$$
is a closed op.

<u>Proof</u>: Let $x_n \to x$ in $N(T)$ as $n \to \infty$. $x \in X$.

$Tx_n = 0 \quad \forall n.$

$Tx_n \to 0$ as $n \to \infty$

From the defn of closed op.

$x_n \to x$

$Tx_n \to y = 0$

$\Rightarrow \quad x \in X_0 \;\; \& \;\; Tx = y.$

$\Rightarrow \quad Tx = 0$

$\Rightarrow \; x \in N(T)$

$\therefore \; N(T)$ is closed.

Let $y_n \in R(T)$ $\;\; y_n \to y$ in Y.

$\& \;\; T^{-1} y_n \to x$ in X.

Define $x_n = T^{-1} y_n$

$Tx_n \to y$ in Y.

& $x_n \to x$ in X

Now T is closed

$\Rightarrow \quad Tx = y$; , $y \in R(T)$.

$\Rightarrow \quad T^{-1}x = y$ & $y \in R(T)$

$\Rightarrow T^{-1}$ is closed.

<u>Ex.</u> Suppose X is a Banach space & $T: X \to Y$ is an injective closed operator.

If $R(T)$ is not closed in Y.

Then what can you say about $T^{-1} : R(T) \subseteq Y \longrightarrow X$?

—

#

T is bdd $\Longleftrightarrow$ for any bdd set E, $T(E)$ is bdd.

#

$$T : X \xrightarrow{\text{bdd}} Y$$

$\{ Tx \mid x \in X \mid \|x\| \leq 1 \}$ is bdd

$\Rightarrow \overline{\{Tx \mid x \in X \mid \|x\| \leq 1\}}$ is bdd.

Also $\overline{\{Tx \mid x \in X \mid \|x\| \leq 1\}}$ is closed. + bdd.

Defn. A linear operator $A : X \to Y$ is compact if $\overline{\{Ax \mid x \in X, \|x\| \leq 1\}}$ is compact.

Suppose T is compact.

$$\overline{B} = \overline{\{Tx \mid x \in X, \|x\| \le 1\}}$$

is ~~bdd~~. compact

$\Rightarrow$ $\overline{B}$ is closed & bdd

$\Rightarrow$ $\overline{B}$ is bdd

$\Rightarrow$ B is bdd.

$\Rightarrow$ $\{Tx \mid x \in X, \|x\| \le 1\}$ is bdd

characterisation

$\Rightarrow$ T is bdd.

$I : X \to X$

Qn: Is I compact?

$$\overline{\{I(x) : x \in X \ \ \|x\| \leq 1\}}$$

$$= \overline{\{\underset{\in X}{x} : \|x\| \leq 1\}}$$

$$= \{x \in X \mid \|x\| \leq 1\} \text{ is compact}$$

$\iff X$ is finite dim.

② Every bdd op of finite rank is compact.

Let $T : X \to Y$ be bdd & $R(T)$ is finite dimensional.

Then $\overline{\{ Tx \mid x \in X, \|x\| \le 1 \}}$ is compact

$\subseteq R(T)$ ($\because T$ is bdd)

$\therefore$ T is compact.

Characterisations of Compact ops.

Theo. Let X, Y be NLS and $T: X \to Y$ be a linear op.

Then the following are equivalent,

i) T is compact.

2) For every bdd E in X $\overline{T(E)}$ is compact.

3) For every bdd seq (x_n) in X $(T(x_n))$ has

a convergent subseq.

(2) $\Rightarrow$ (1) clearly.

(1) $\Rightarrow$ (2)

Assume T is compact.

ie $\overline{\{Tx \mid x \in X, \|x\| \leq 1\}}$ is a compact subset of Y.

Let $E \subseteq X$ be a bdd set.

$\exists\ r > 0$ s.t

$$E \subseteq \{x \in X, \|x\| \leq r\}.$$

$$\therefore \quad T(E) \subseteq \{Tx \mid \|x\| \le r\}.$$

$$\Rightarrow \quad \overline{T(E)} \subseteq \overline{\{Tx \mid \|x\| \le r\}}$$

If $\overline{\{Tx \mid \|x\| \le r\}}$ is compact

then $\overline{T(E)}$ being the closed subspace will be compact

Now we have

$$\overline{\{Tx \mid \|x\| \le 1\}} \text{ is compact}$$

$$\Longleftrightarrow \quad \overline{\{ Tx \mid \|x\| \leq r \}} \text{ is compact}$$

(Ex).

(1) $\Leftrightarrow$ (2)

(2) $\Rightarrow$ (3).